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(19) **United States**(12) **Patent Application Publication****Betts et al.**(10) **Pub. No.: US 2010/0080082 A1**(43) **Pub. Date: Apr. 1, 2010**(54) **SIDE SCAN SONAR IMAGING SYSTEM**

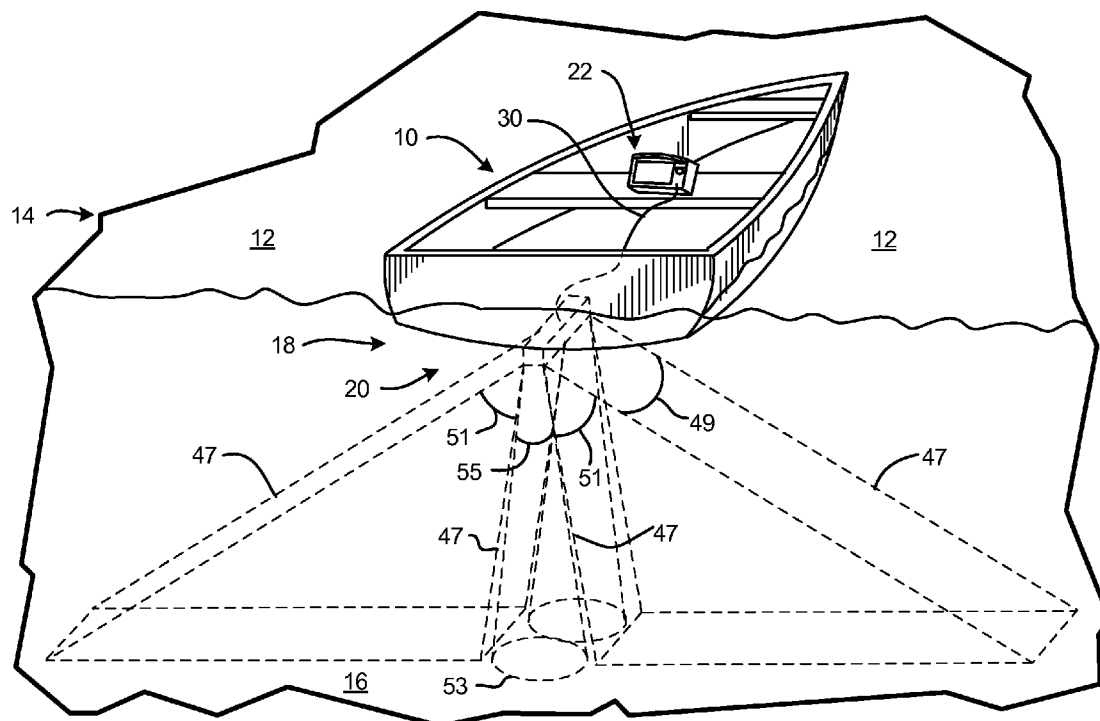
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(75) Inventors: **David A. Betts**, Eufaula, AL (US);
Robert W. Derrow, Eufaula, AL
(US); **David J. Howells**, Atlanta,
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Correspondence Address:

KINNEY & LANGE, P.A.**THE KINNEY & LANGE BUILDING, 312
SOUTH THIRD STREET
MINNEAPOLIS, MN 55415-1002 (US)**(57) **ABSTRACT**(73) Assignee: **JOHNSON OUTDOORS INC.**,
Racine, WI (US)(21) Appl. No.: **12/631,229**(22) Filed: **Dec. 4, 2009****Related U.S. Application Data**(63) Continuation of application No. 11/195,107, filed on
Aug. 2, 2005, now Pat. No. 7,652,952.

A system for use with a boat to provide underwater sonar images includes a GPS receiver for providing GPS position data, a left side scan sonar transducer for transmitting left side scan sonar pulses and for receiving left side scan sonar return signals, and a right side scan sonar transducer for transmitting right side scan sonar pulses and for receiving right side scan sonar return signals. The system further includes signal processing circuitry for processing the left and right side scan sonar return signals to produce side scan image data and a digital processor for causing a display to display an underwater image based upon the side scan image data, wherein the digital processor associates GPS position and side scan image data.



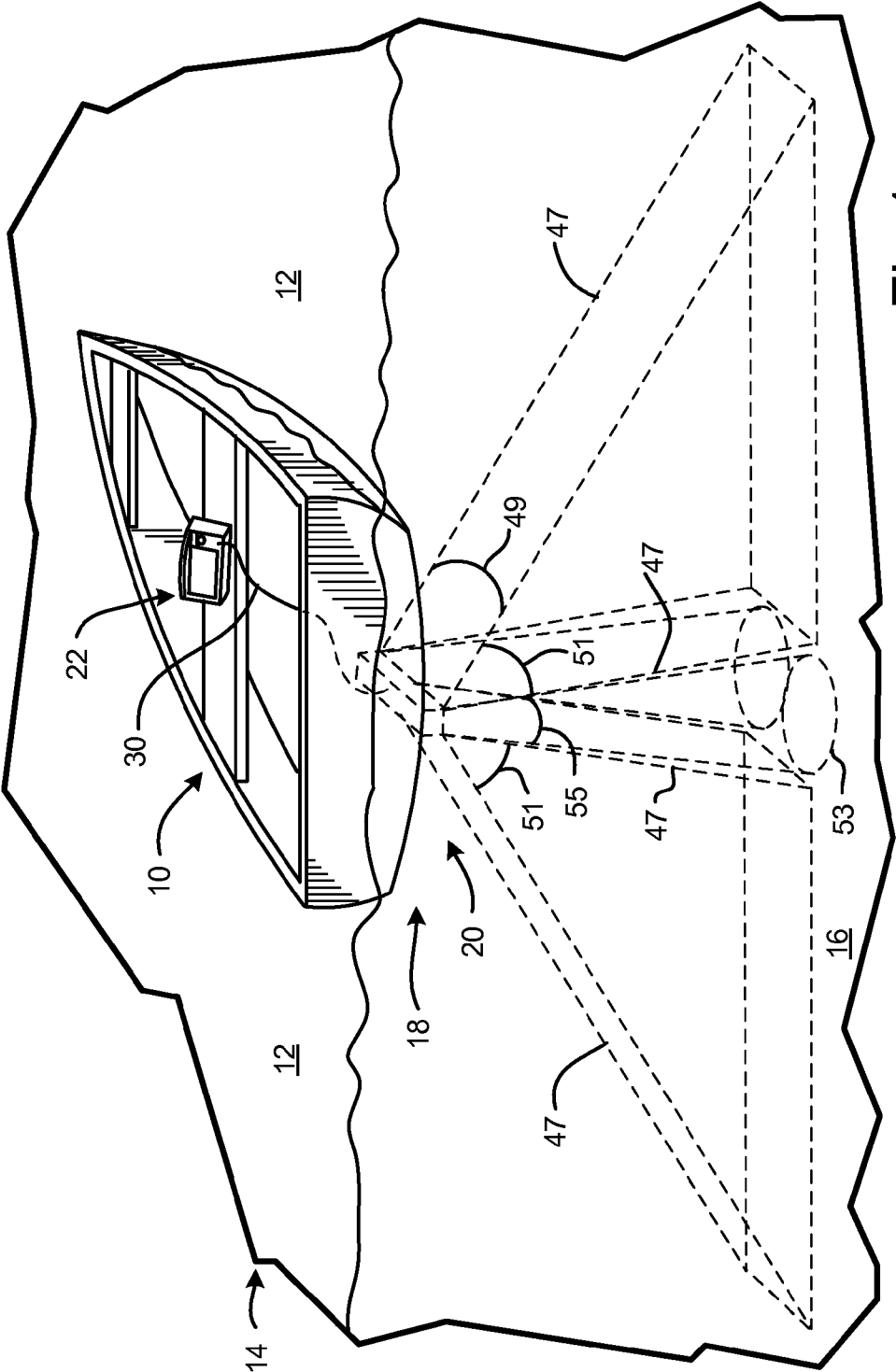


Fig.1

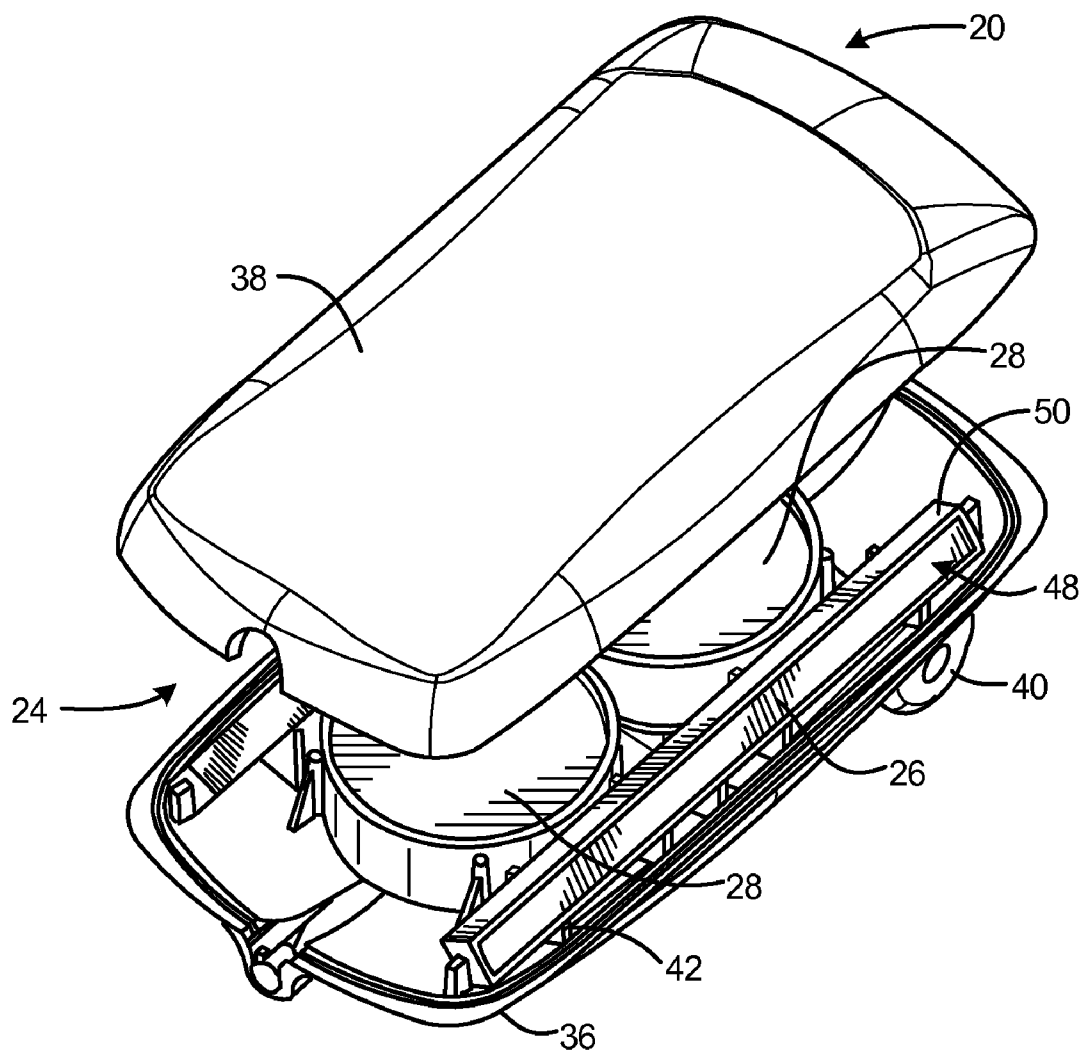
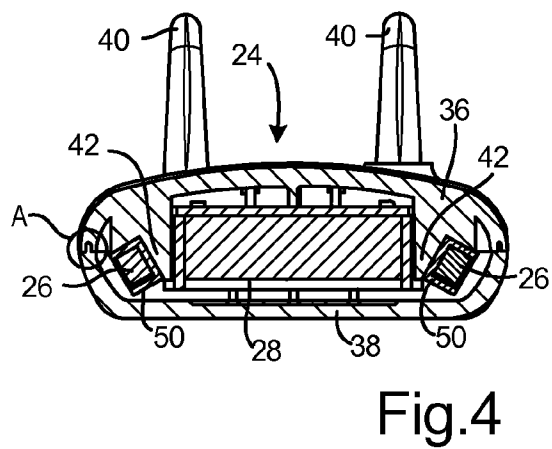
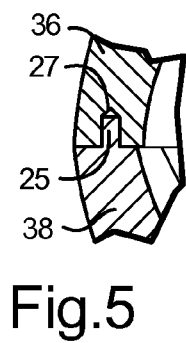
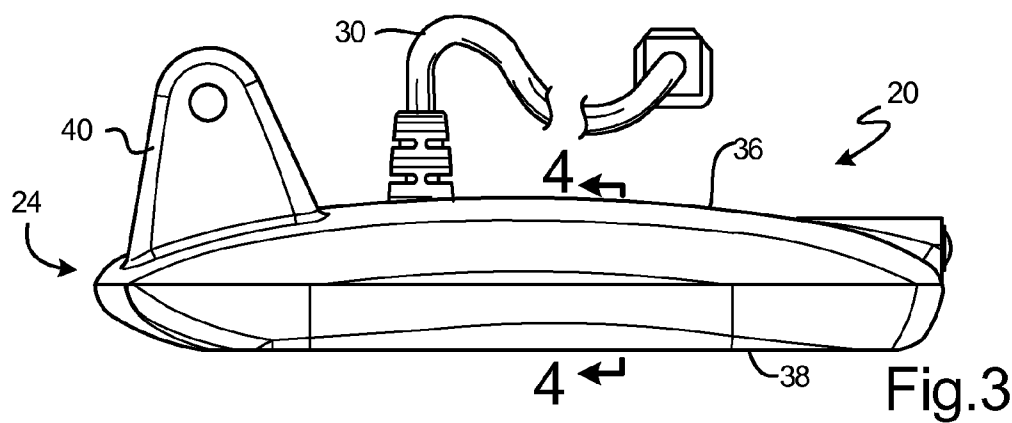


Fig.2



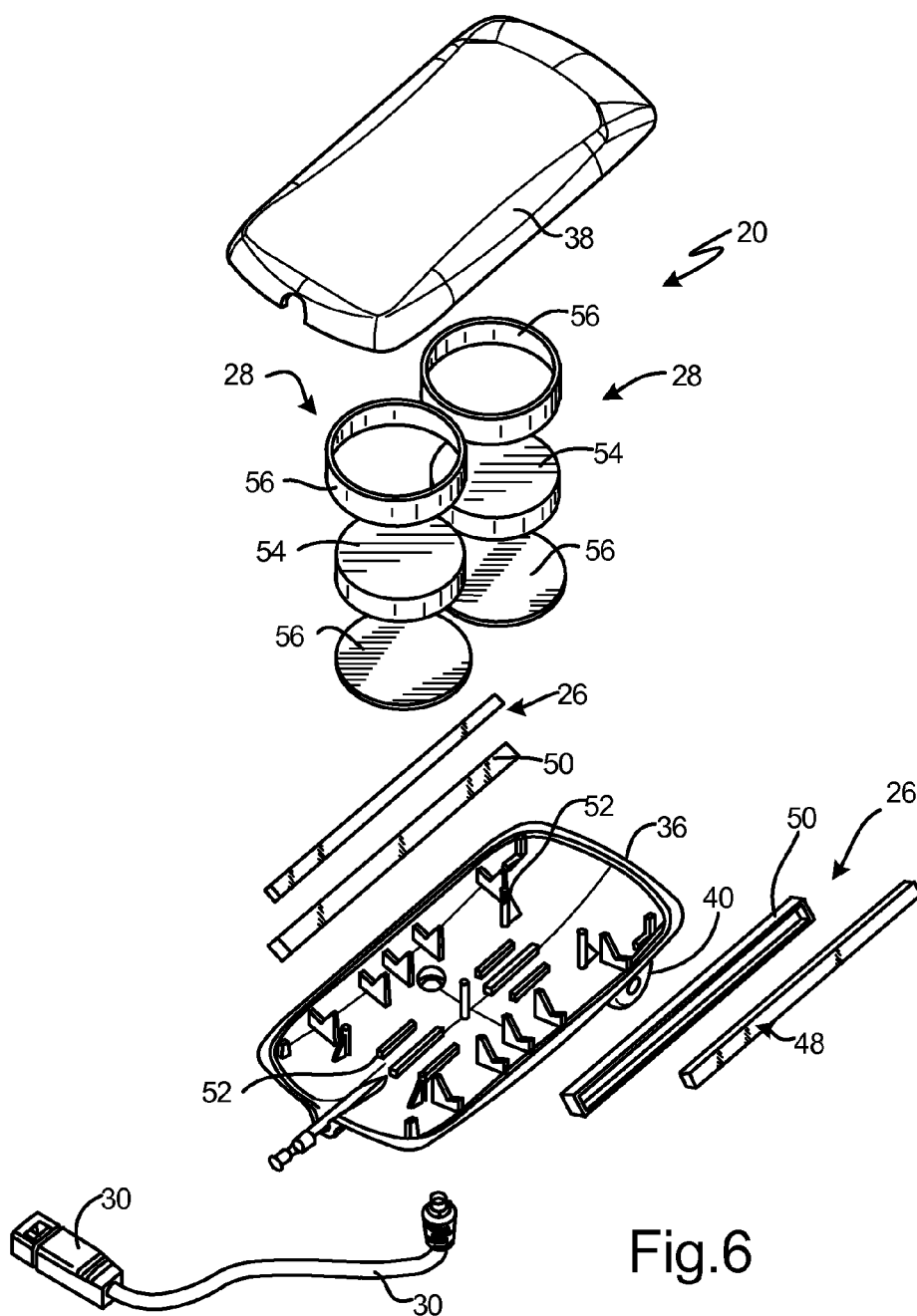


Fig.6

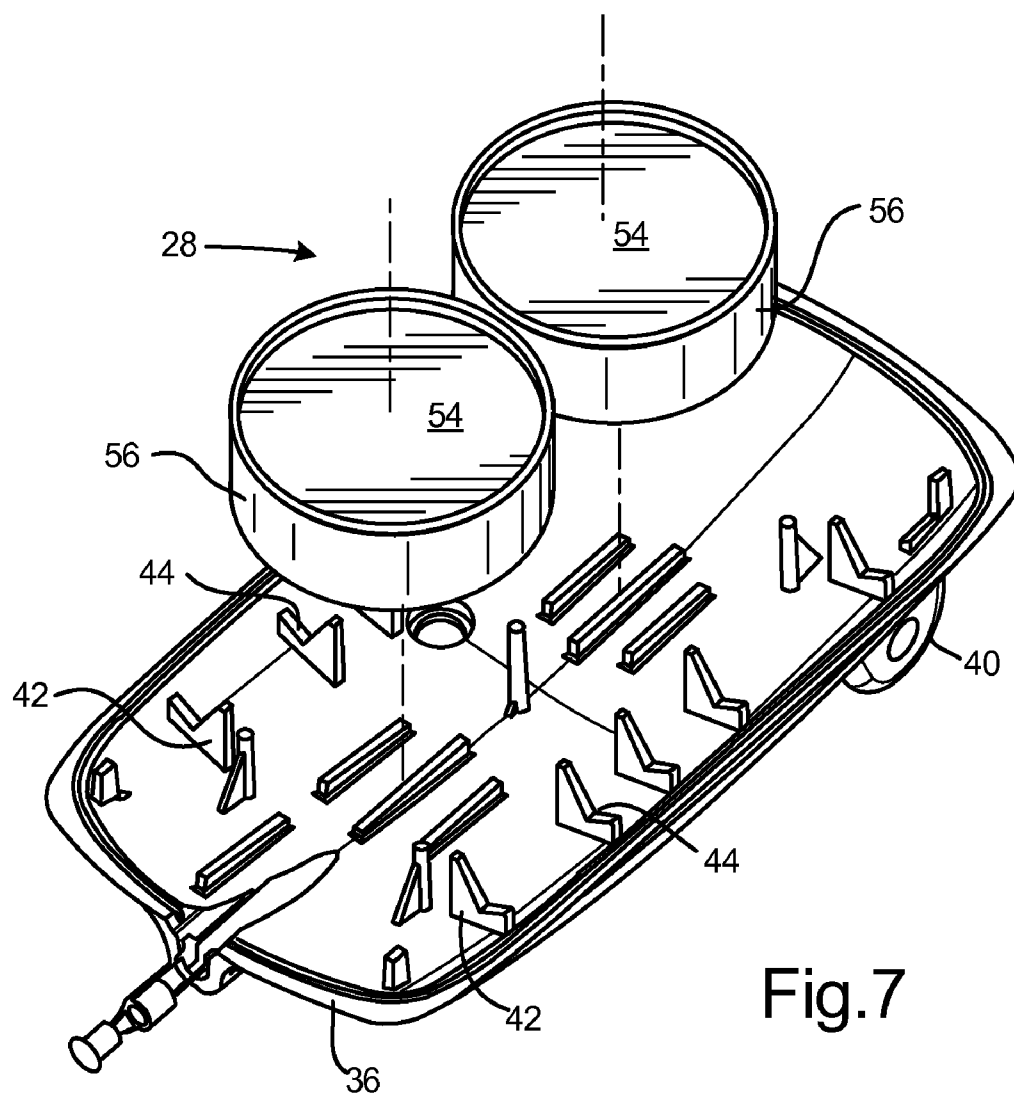
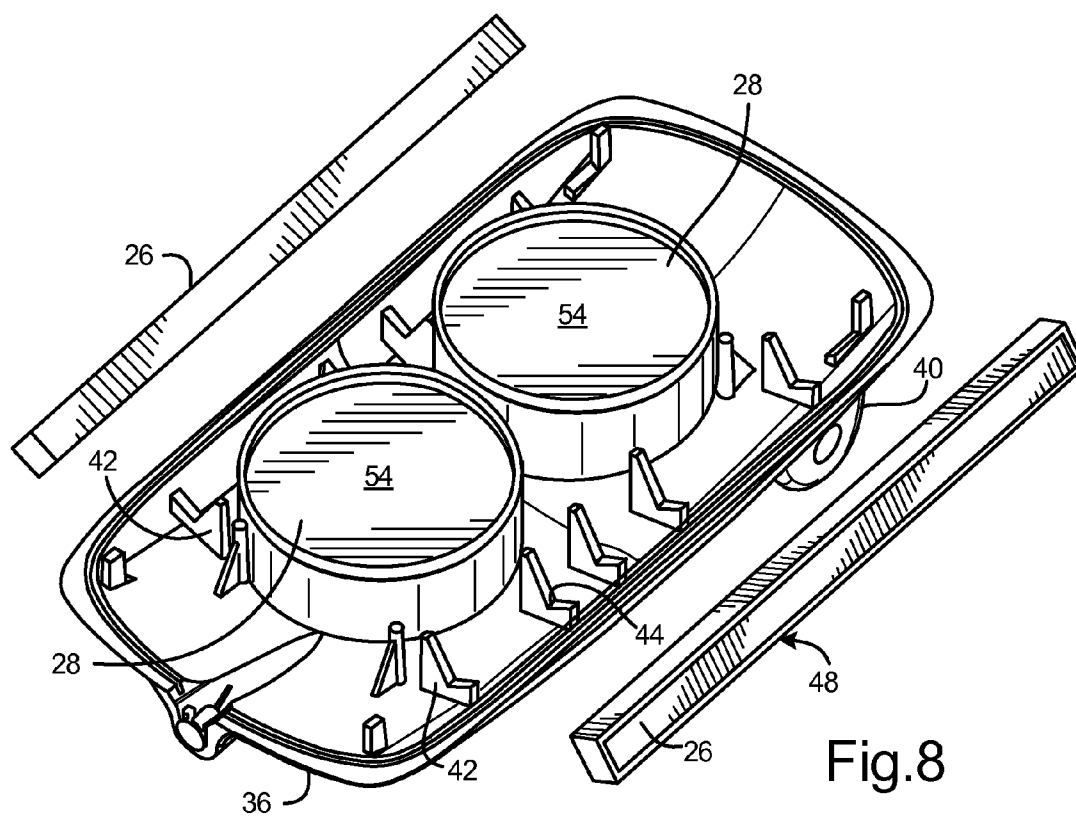


Fig.7



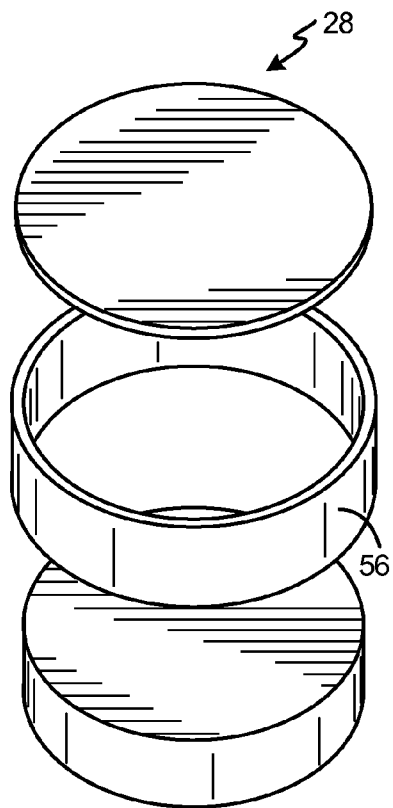


Fig.9

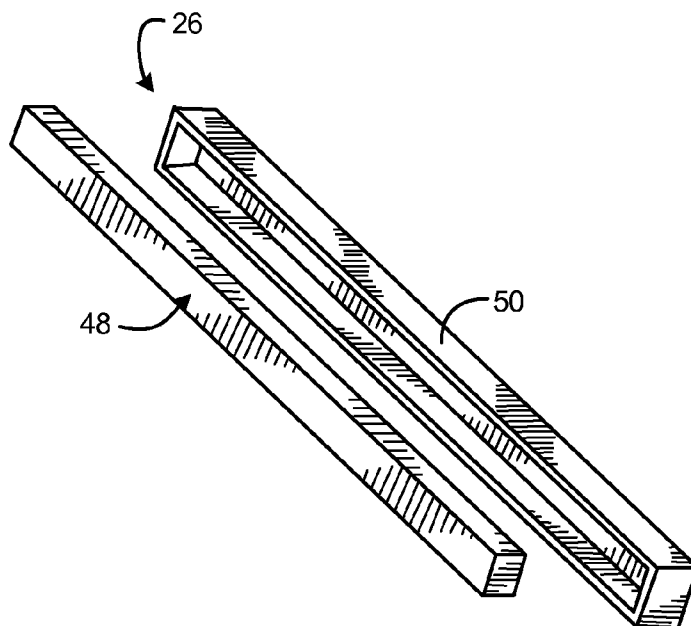


Fig.10

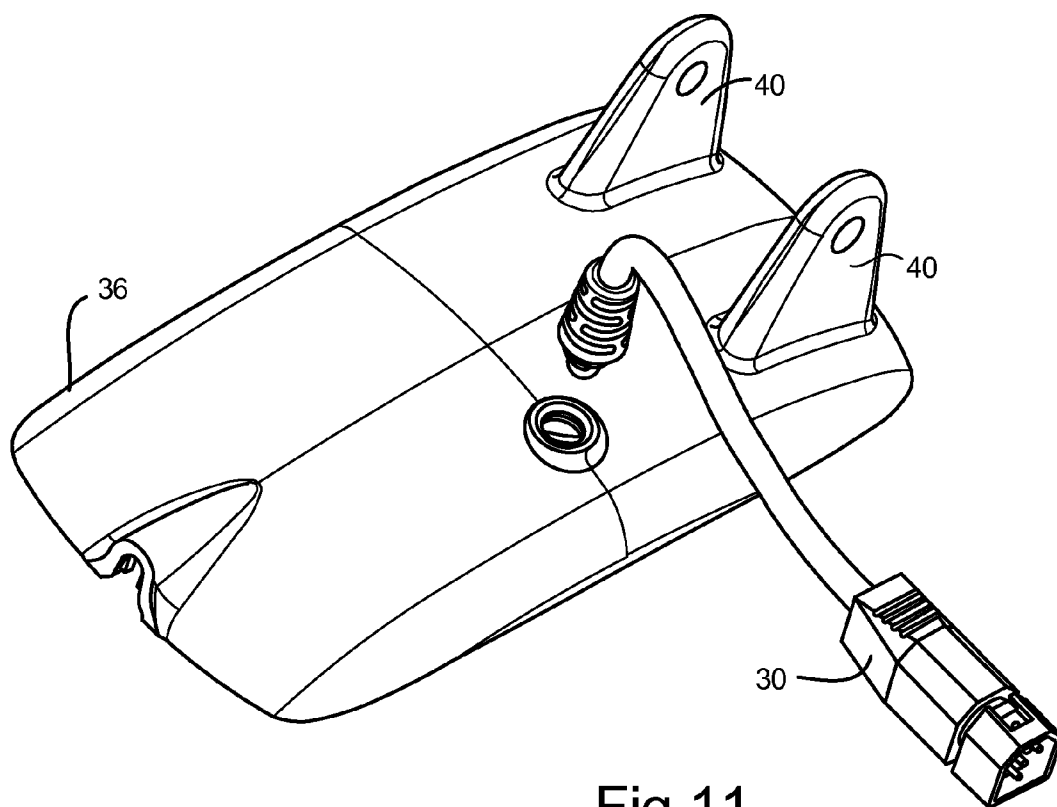


Fig.11

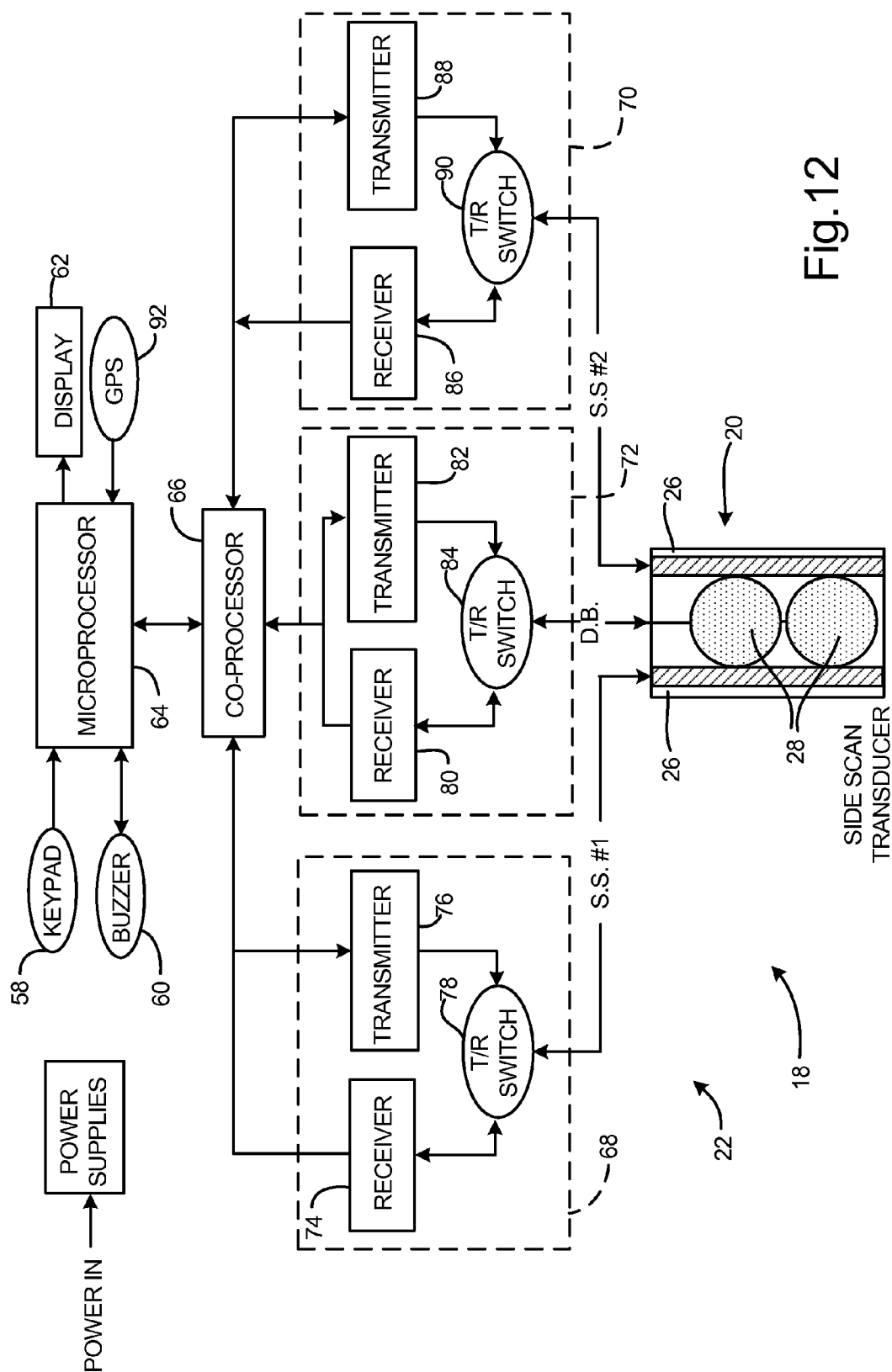


Fig.12

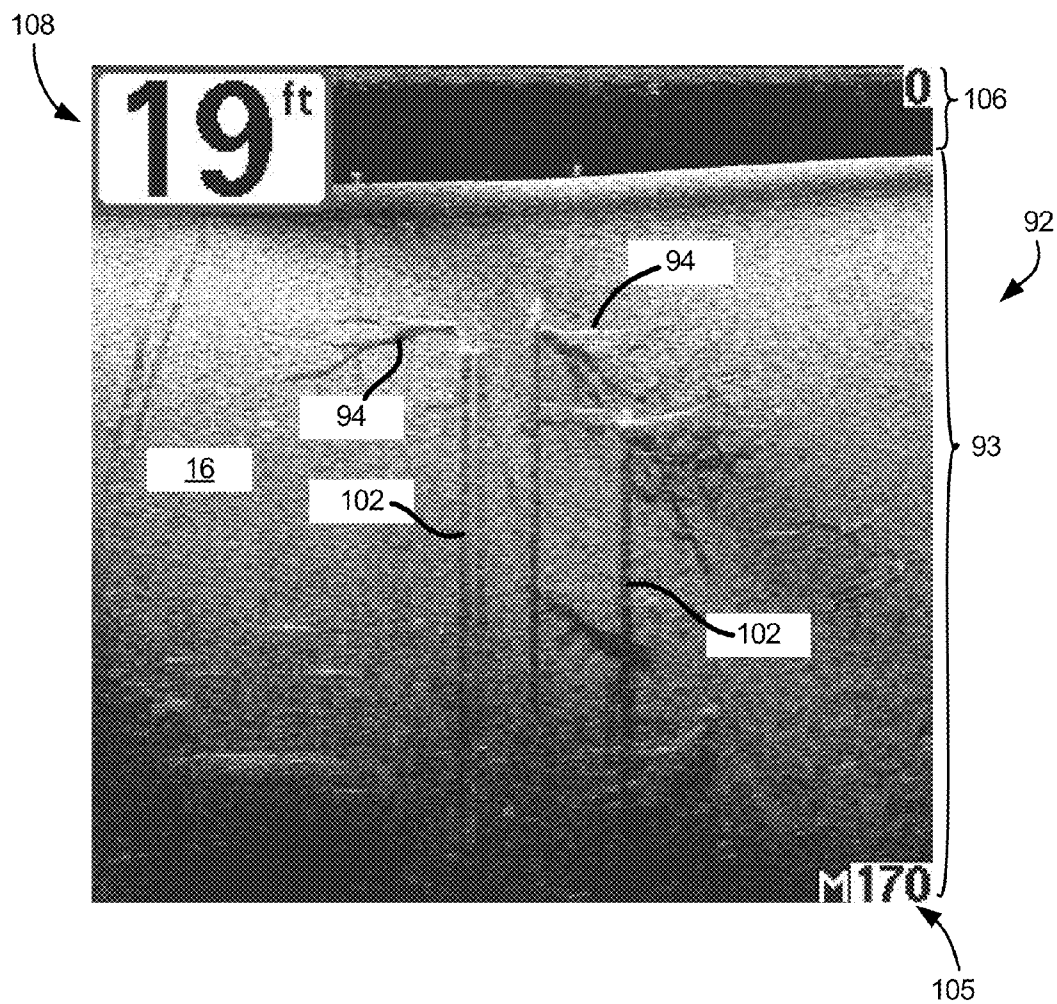


Fig. 13

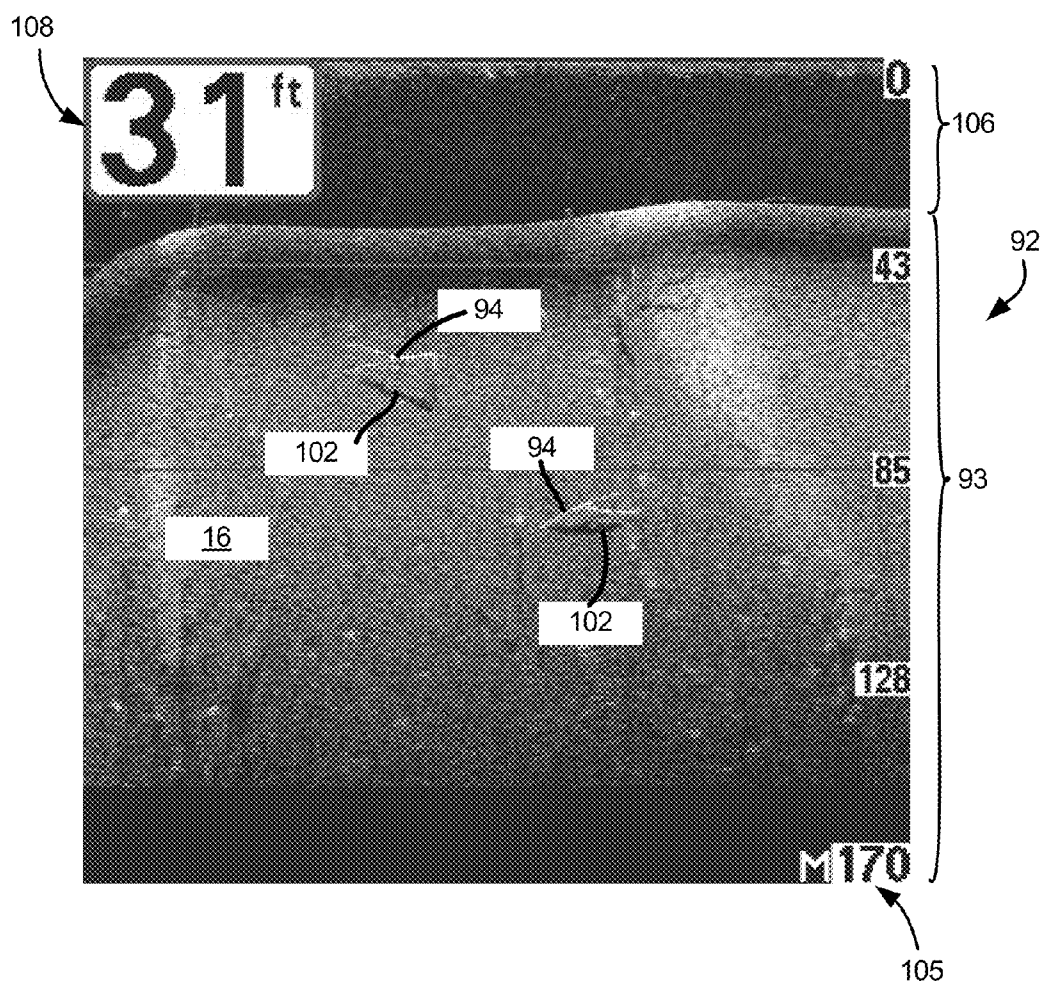


Fig. 14

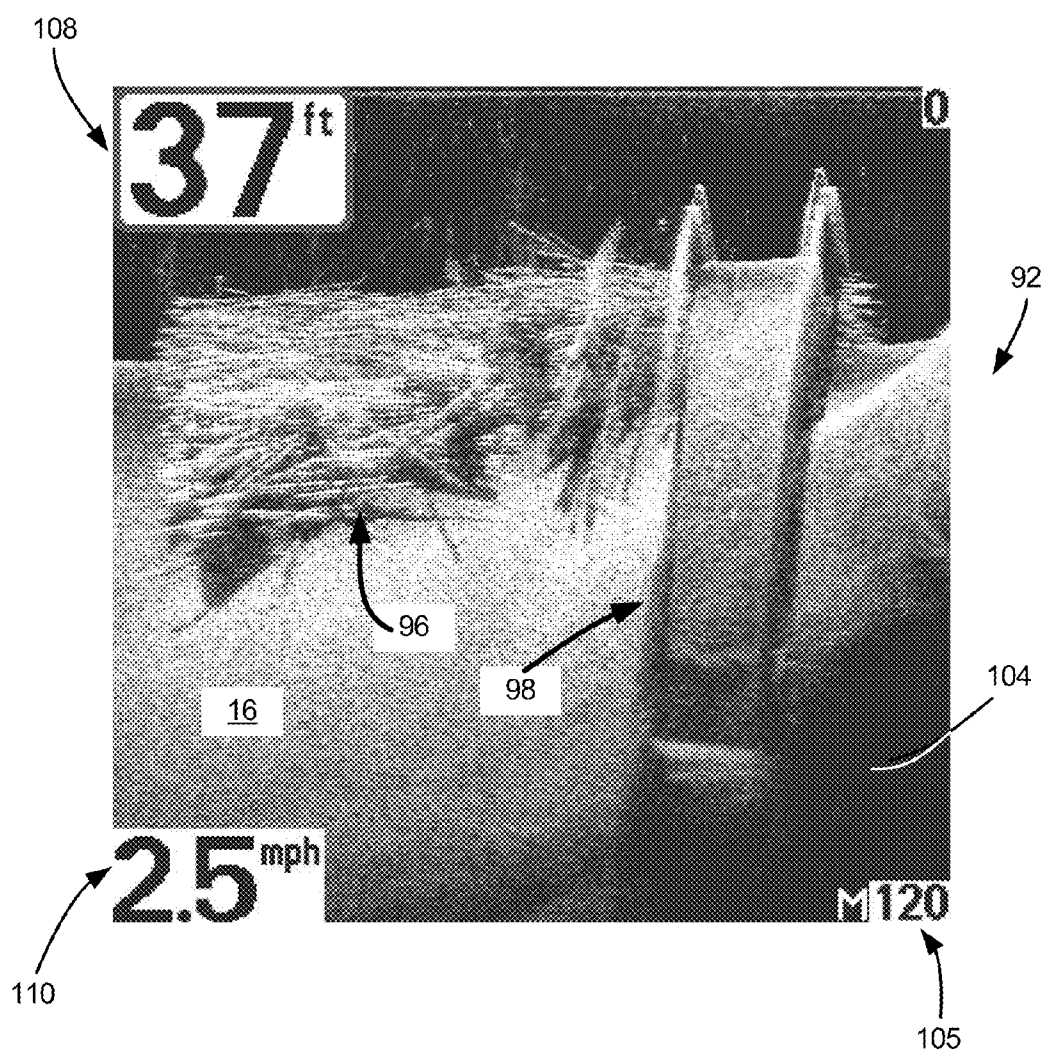


Fig. 15

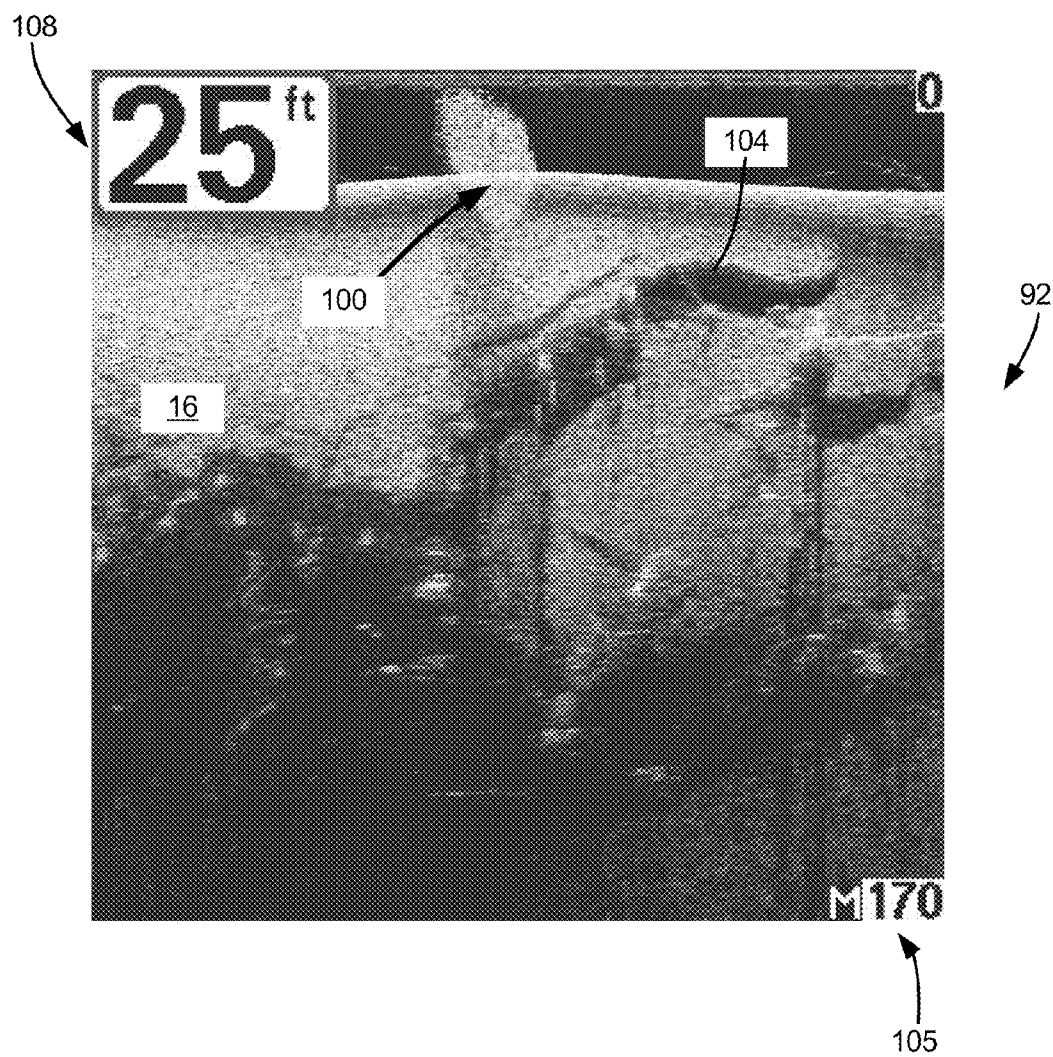


Fig. 16

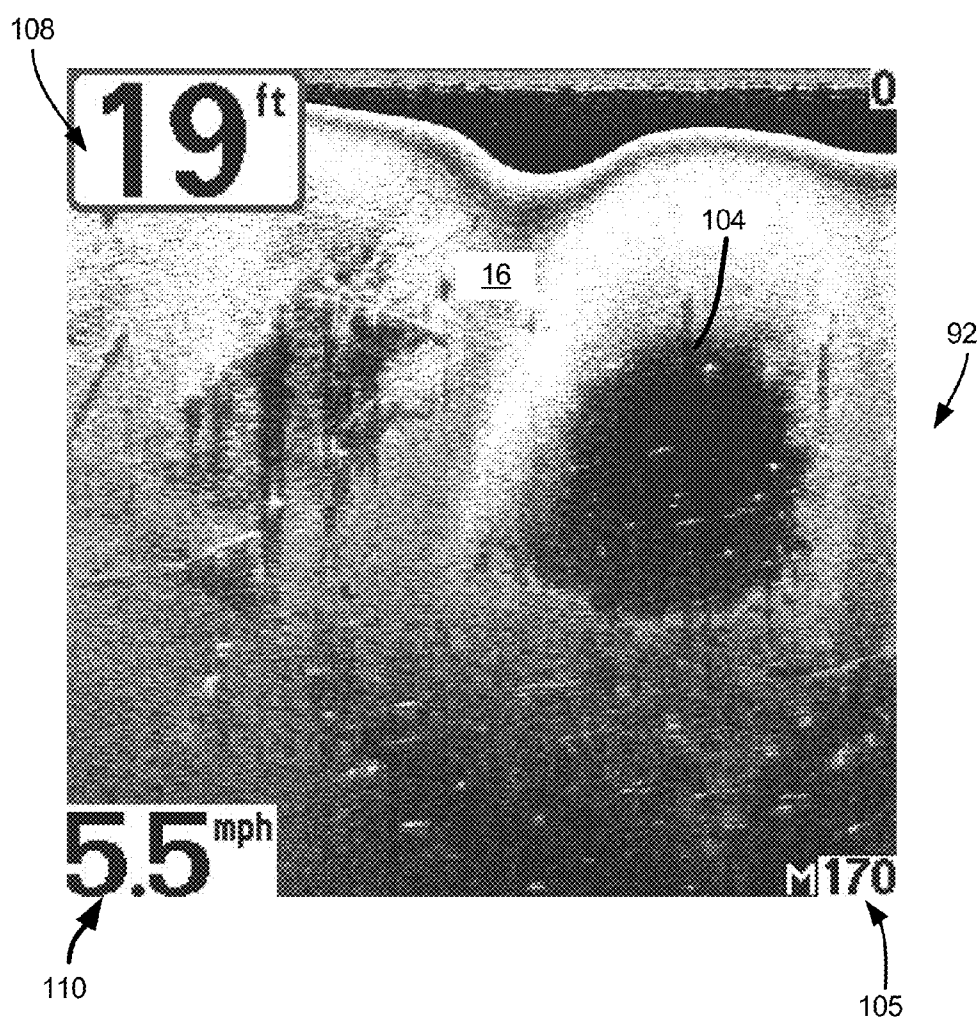


Fig. 17

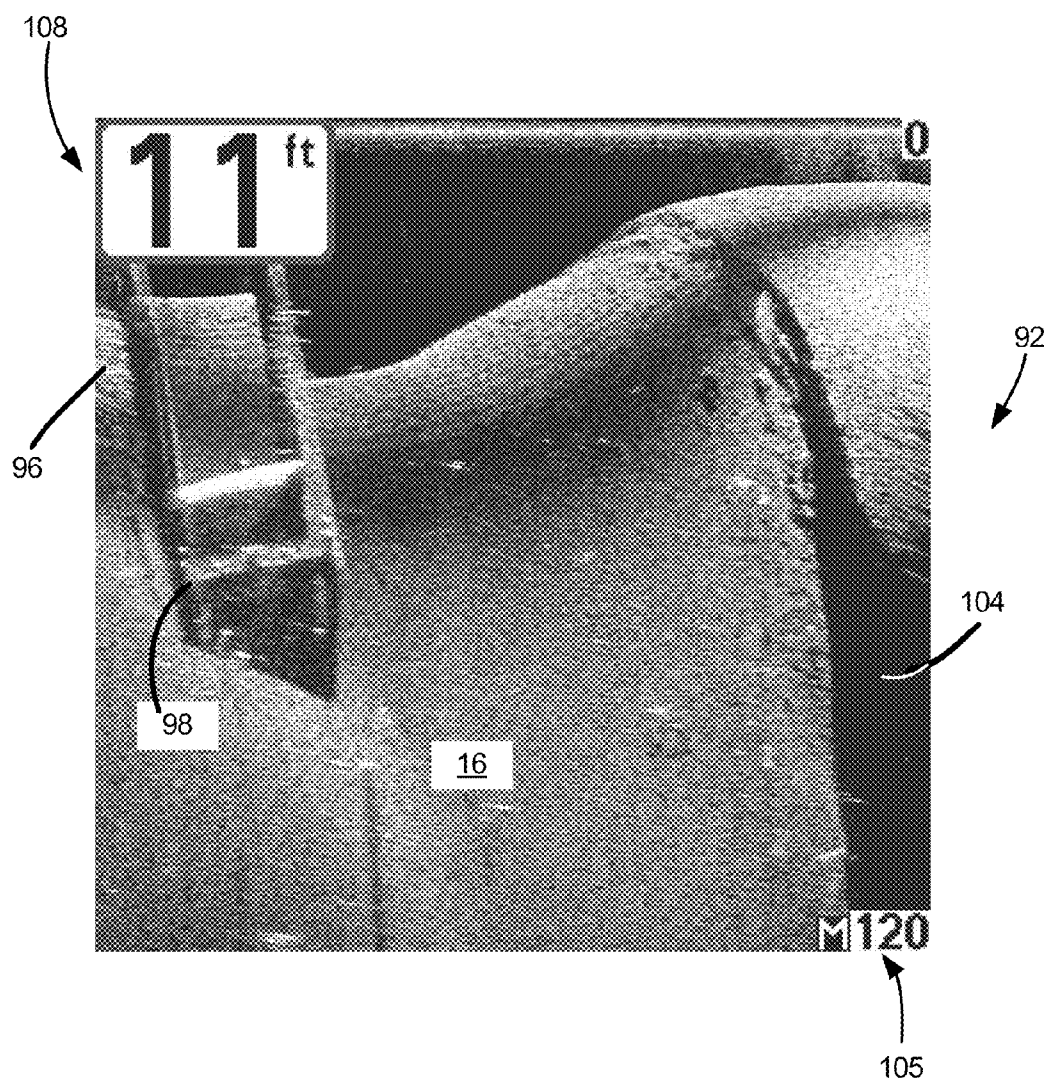


Fig. 18

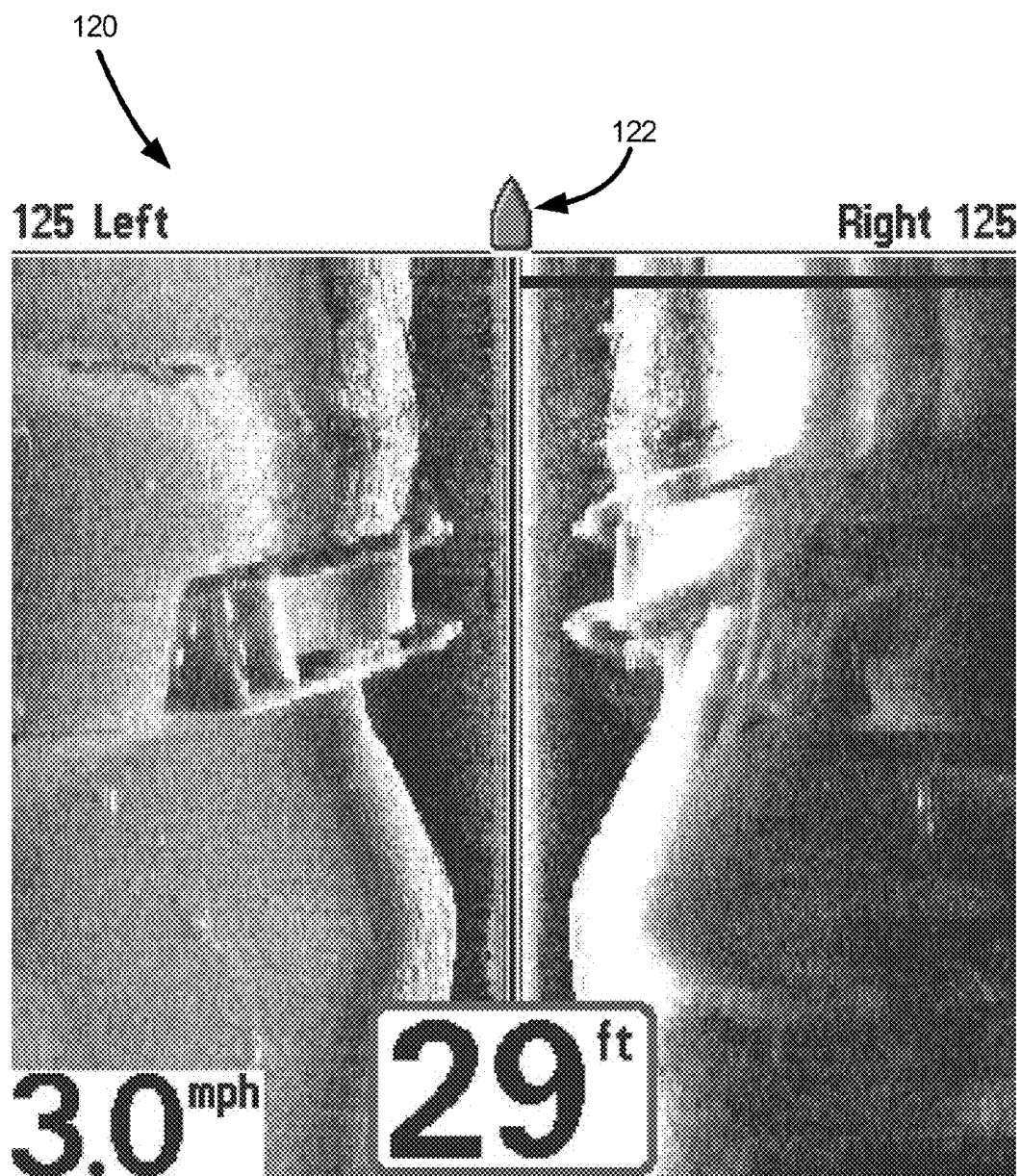


Fig. 19

SIDE SCAN SONAR IMAGING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This is a continuation of U.S. patent application Ser. No. 11/195,107, entitled "SONAR IMAGING SYSTEM FOR MOUNTING TO WATERCRAFT," filed Aug. 2, 2005 by David A. Betts et al. and claims the benefit of U.S. Provisional Patent Application No. 60/598,326, filed Aug. 2, 2004, the teachings and disclosure of which are hereby incorporated in their entirety by reference thereto.

[0002] Reference is also made to application Ser. No. 12/319,594 entitled "SIDE SCAN SONAR IMAGING SYSTEM WITH BOAT POSITION ON DISPLAY", application Ser. No. 12/319,586 entitled "SIDE SCAN SONAR IMAGING SYSTEM WITH IMAGE ENHANCEMENT", and application Ser. No. 12/319,604 entitled "SIDE SCAN IMAGING SYSTEM WITH ASSOCIATED GPS DATA", which are also continuations of application Ser. No. 11/195,107, and which were filed on Jan. 9, 2009 and are assigned to the same assignee as this application.

FIELD OF THE INVENTION

[0003] The present invention relates generally to sonar imaging systems for use in sport fishing applications such as in a fish finder, sonar depth sounder, etc., and more particularly to side scan sonar imaging systems for imaging of the underwater environment to the sides of the watercraft rather than just below the watercraft.

BACKGROUND OF THE INVENTION

[0004] Sonar devices that transmit sound waves have been used previously to obtain information about underwater articles, including fish, structures and obstructions, and the bottom. The sound waves travel from a transducer mounted to a bottom surface of the vessel through the water. The sound wave transmits from the sonar devices in diverging patterns. The sound waves contact underwater articles, which create return echoes. The transducer receives the return echoes and the sonar device analyzes the received echoes. A display device displays representations of the received echoes, for locating fish and other underwater articles.

[0005] Known side scan sonar devices locate the transducer in a vessel towed by the watercraft (e.g., a "tow fish"). The tow fish is coupled to the sonar display by a long cable. The length of the cable will depend on the depth of the water and other conditions. For typical applications, the length of the cable is 50 feet or more. Moreover, it is not uncommon for the cable to be hundreds or even thousands of feet long. As can be appreciated by some having ordinary skill in the art, a fisherman or recreation user desiring to have side scan images would be hindered by such an arrangement. For example, maneuvering or turning of the watercraft in different directions is difficult, as well as tangling of the sonar cable with fishing or other recreational equipment. Such known tow fish transducers are maintained at a consistent distance from the bottom of the body of water. This distance is intended to provide desired or optimized resolution and field of view. A consistent distance inhibits, if not prohibits, modifying known transducers for side scan applications (or mounting known side scan transducers to watercraft) because the dis-

tance between the transducer to the bottom of the water will vary as the watercraft travels due to the varying depth of the water.

[0006] Accordingly, it would be advantageous to provide a sonar imaging system that is coupled to the watercraft, rather than being coupled by a flexible cable and towed behind the watercraft. It would also be advantageous to provide sonar imaging system mountable to a motor (such as a trolling motor), a transom of the watercraft, or to the hull of the watercraft. It would also be advantageous to provide sonar imaging system operable at multiple resonant frequencies for optimized performance at varying bottom depths. It would be desirable to provide for a sonar imaging system for mounting to a watercraft having one or more of these or other advantageous features.

BRIEF SUMMARY OF THE INVENTION

[0007] In view of the above, it is an objective of the present invention to provide a new and improved sonar imaging system that is capable of being connected to a watercraft, such as a fishing boat. It is a further objective to provide a new and improved sonar imaging system that provides imaging of the underwater environment to the sides of the watercraft. It is a still further object of the invention to provide a new and improved sonar imaging system that additionally provides imaging of the underwater environment below the watercraft.

[0008] The system of the present invention realizes several advantages over the traditional towfish side scan sonar systems. It is more convenient because there are no deployment requirements of getting the transducer towfish into the water, no cable handling hassles and tangles, no precise speed control requirements to keep the towfish at the right depth and prevent it from hitting the bottom, no complicated large diameter turn requirements to prevent the towfish from hitting the bottom when you want to turn the boat, and no worries about getting the lines and cable tangled when fishing. The system of the present invention can even be used for imaging by a watercraft in reverse.

[0009] The system of the present invention is also more secure than the traditional side scan sonar systems. There is no chance of snagging the towfish or loss of transducer. Most fishing is done near bottom rises, drop-offs and underwater structures. Most natural and especially man-made lakes have rocks, stumps and standing timber that can snag a towfish and cause damage or loss of the equipment.

[0010] Additionally, the system of the present invention provides more area of coverage. The watercraft mounted transducer of the present invention does not limit the turning radius of the vessel, and it provides the ability to image closer to the shore and near structure. This allows for faster and more complete imaging. The system also provides more accurate target locations. Having the transducer mounted to the watercraft allows for precise target locations. With a towed side scan system the crew has to take into account how much cable is deployed and how deep the towfish is to determine how far back behind the boat the target is. With the watercraft mounted system of the present invention, this is not a factor. To provide even more accurate images, the system of the present invention provides the offset necessary to account for the X and Y distance between the side imaging transducer and the GPS antenna. The system of the present invention also has a better aspect at some targets because, in some cases, the view from the surface can "see" better over rises and into holes than the towed side scan sonar at a fixed distance from

the bottom. The system of the present invention can also be mounted to smaller watercraft such as canoes, kayaks and other personal watercraft.

[0011] In a preferred embodiment of the present invention, the system includes features to correct for watercraft mounted nature of the transducers. Unlike using a towfish in which data collection takes place at a fixed distance from the bottom (same aspect angle at any depth of water), and in which the towfish dynamics are decoupled from vessel motion in rough seas, the system of the present invention compensates for these differences. In a preferred embodiment of the present invention, the depression angle of the side imaging elements is increased from about 20 degrees to about 30 degrees. This provides better coverage at the greater aspects. Also in a preferred embodiment, the side elements are designed to be dual frequency to provide a trade-off between area of coverage and resolution. Transducer element shielding and software filters are also provided in a preferred embodiment to eliminate vessel noise sources such as spark plug and electrical system EM1 (solenoids, VHF radios, electric motors, etc.). In one embodiment of the present invention, the system includes passive yaw, tilt transducer minimization or compensation using floating oil bath self leveling. In another embodiment the system includes active yaw, tilt transducer minimization or compensation via tilt sensors and motors.

[0012] Additional features over traditional side scan provided by embodiments of the present invention include fish identification and alarm in side beams. Typical side scan systems consider fish as unwanted noise. Screen capture and playback functioning like a digital camera with the ability to store an image, review already stored images, erase unwanted images, and download images to a computer are also provided in embodiments of the present invention. Unlike typical data recording when a user sees an image on the screen they can simply push a capture button, instead of having to start recording before the user sees the target. Preferred embodiments also provide zoom capability that allows a user to view only the right or left side at a time and also zoom into a particular area either using the cursor or a touch screen. Further, the ability to use standard image enhancement software (algorithms) either in the unit or post processed is provided to allow for color, contrast, brightness, auto fix, edge detect, etc.

[0013] In a preferred embodiment of the present invention, a down beam is provided along with the side imaging. This provides for more complete around the boat information (both sides and straight down). It is not limited to a single beam. One embodiment utilizes a 200 kHz/50 kHz dual beam. Other embodiments may use a quad beam or even six beam. In preferred embodiments, at least one view shows both down beam and side imaging. This provides the ability to better relate length of shadow information to the size of the underwater target. It also provides for a quick means for verification of target location. After a target is located off to a side, the boat can be driven directly over the target and located in the down beam for precise location.

[0014] In a still further preferred embodiment, GPS imaging is also provided with the side imaging. In such embodiments a cursor mode allows a user to move the cursor over a target of interest on the screen image and set a waypoint for the location of the structure. The GPS history may be used to determine the distance back and the sonar may be used to determine the distance to the side. The GPS speed can be used to provide the screen scroll rate to provide more accurate front

to back target dimensions. Without GPS or a speed sensor a fast scroll rate and a slow boat speed will elongate targets and a slow scroll rate and a fast boat speed will shorten targets. The corners of screen captures can be marked so that large area composite mosaic images can be generated in the unit or post processed later. Preferably, one view that shows both side imaging and navigation information is provided. This makes it easier to follow tracks and provide efficient area coverage.

[0015] In accordance with these objectives, an embodiment of the present invention provides a sonar imaging system for a watercraft including a transducer coupled to the watercraft. Preferably, the system includes at least one side scanning element and at least one bottom scanning element and an electronic control head unit coupled to the transducer that is configured to display sonar images. In one embodiment of the present invention, the sonar imaging system includes circular downward acoustic elements and rectangular acoustic elements. The present invention further relates to a software filter configured to remove noise generated by a spark plug or other operation of a motor for the watercraft.

[0016] Other aspects, objectives and advantages of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The accompanying drawings incorporated in and forming a part of the specification illustrate several aspects of the present invention and, together with the description, serve to explain the principles of the invention. In the drawings:

[0018] FIG. 1 is an isometric illustration of one embodiment of a fishing vessel mounted sonar imaging system of the present invention.

[0019] FIG. 2 is an exploded bottom view isometric illustration of an embodiment of a transducer module constructed in accordance with the teachings of the present invention.

[0020] FIG. 3 is a side view illustration of the assembled transducer module of FIG. 2.

[0021] FIG. 4 is an end section view of the assembled transducer module of FIG. 3 taken about section line 4-4.

[0022] FIG. 5 is a partial section view of the housing joint of the assembled transducer module of FIG. 4 taken at section A.

[0023] FIG. 6 is a fully exploded bottom view isometric illustration to the transducer module of FIG. 2.

[0024] FIG. 7 is a partial exploded view isometric illustration of a top housing assembly showing placement of downward looking sonar elements.

[0025] FIG. 8 is a partial exploded view isometric illustration of a top housing assembly showing placement of downward looking sonar elements and side scan sonar elements.

[0026] FIG. 9 is an exploded isometric illustration of an embodiment of a downward looking sonar element suitable for application in the sonar imaging system of the present invention.

[0027] FIG. 10 is an exploded isometric illustration of an embodiment of a side scan sonar element suitable for application in the sonar imaging system of the present invention.

[0028] FIG. 11 is an isometric illustration of one embodiment of a cable attachment for the sonar imaging system of the present invention.

[0029] FIG. 12 is a simplified system block diagram of an embodiment of the sonar imaging system of the present invention.

[0030] FIGS. 13-18 show examples of sonar images as they appear on a display screen.

[0031] FIG. 19 shows a display screen showing an underwater environment using sonar images from both side scan elements and sonar returns from a downward sonar element.

[0032] While the invention will be described in connection with certain preferred embodiments, there is no intent to limit it to those embodiments. On the contrary, the intent is to cover all alternatives, modifications and equivalents as included within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE INVENTION

[0033] Turning now to the drawings, FIG. 1 illustrates a vessel (shown as a watercraft 10) on a surface 12 of a body of water 14 having a bottom 16. A sonar imaging system 18 is mounted or coupled to the watercraft 10 (e.g., rather than being towed by a flexible cable behind the watercraft 10) and is configured to scan the water below and to the sides of the watercraft (i.e., a boat mounted side scan sonar system). The sonar imaging system 18 comprises a transponder or transducer 20 coupled to an electronic control head unit 22 located at the watercraft 10. The sonar imaging system 18 repetitively scans the body of water 14 for fish and other underwater articles with transmissions of acoustic waves and receiving and displaying the sonar returns, with the duration of receiving being a function of the determined depth from a prior transmission.

[0034] Referring to FIGS. 2-11, the transducer 20 includes a housing 24, a sonar array (in the form of a plurality of acoustic elements shown as side scan elements 26 and downward scan element 28), a cable 30 coupling the housing and acoustic elements to the electronic control head unit 22. The acoustic elements are configured for acoustic wave transmitting and receiving by a transmitter and a receiver that are operated by the electronic control head unit 22 for scanning the body of water 14, particularly for locating fish, as well as other underwater articles, and determining characteristics about the bottom 16 (see FIG. 1) of the body of water. The acoustic or sonar wave beams are transmitted based on the configuration of the acoustic elements.

[0035] The housing 24 comprises a top housing portion 36 and a bottom housing portion 38. The top housing portion 36 includes a pair of mounting members 40 extending from the front portion of the top housing portion 36. Both top housing portion 36 and bottom housing portion 38 have projections extending towards the interior of the housing to provide structural support for the housing assembly, and to provide locating and positioning support for the acoustic elements. A series of projections 42 include v-shaped recesses or notches 44 to form a cradle that receives side scan elements 26. Recesses 44 are configured (shaped and positioned) to support the rectangular shaped side scan elements 26 in a position and orientation (direction) to provide a particular, desired, predetermined acoustic beam performance.

[0036] The housing 24 is coupled to the watercraft 10 by any of a variety of methods. The housing 24 is coupled to the watercraft 10 so that there are no obstructions to either side of the housing (i.e., to block the operation or affect the performance of the acoustic elements). According to a preferred embodiment, the housing 24 is coupled to the watercraft 10 along the centerline of the watercraft so that the housing 24 extends about 0.25 inches below the watercraft. According to an exemplary embodiment, mounting members 40 of the top

housing 36 are coupled to a mounting bracket 46 that is coupled to a trolling motor. According to an alternative embodiment, the mounting members 40 are mounted through the hull of the watercraft 10 (e.g., with a support shaft passing through a hole in the hull). According to an alternative embodiment, the mounting members are coupled to a bracket that is coupled to a transom of the watercraft 10. Alternatively, the housing may be coupled to the watercraft at any of a variety of positions and at any of a variety of depths below the surface of the water. The top housing portion 36 may be coupled to the bottom housing portion 38 by any of a variety of conventional methods (e.g., snap fit engagement, adhesive, ultrasonic welding, fusion, heat welding, fasteners such as screws, bolts, rivets, or the like). As illustrated in FIG. 5, one embodiment of the housing 24 utilizes a mounting rib 25 that is received in a mounting channel 27 to locate the top and bottom housing portions 36, 37 together. In one embodiment, the rib 25 is welded into the channel 27 such as, e.g. via ultrasonic welding or the like.

[0037] The side scan elements 26 are located along the sides of the housing 24 and are configured to scan the water to the sides of the transducer 20 (and watercraft 10) with sonar or acoustic beams 47.

[0038] The dimensions of side scan elements 26 are configured to provide the desired sonar beam pulse 47. The size of the wave front created by the transmitted acoustic beam affects the resolution of the return echo and thus the quality of the imaging of subsurface articles displayed by the sonar device. Generally, a wide beam provides diffused return echoes that are particularly suited for indicating the presence of fish in a wide area around the watercraft. The signal displayed for fish is referred to as a "fish arch" or other indicia or icon. A narrower beam on the other hand provides a more detailed return echo or signal representative of the subsurface article. The narrow beam covers a smaller area but provides additional definition of the article. A wider beam accordingly is useful for providing indications of the presence of schools of fish in a wide area around the vessel as well as other underwater articles. The narrow beam is useful for providing details of the underwater article or the bottom.

[0039] A sonar beam becomes narrower (thereby providing better resolution) as the corresponding dimension of the acoustic element becomes larger, and a sonar beam becomes wider as the corresponding dimension of the acoustic element becomes smaller (e.g., a small height provides for a beam with a relatively wide vertical angle, and a large length provides for a beam with a relatively narrow horizontal angle). According to an exemplary embodiment, the side scan elements 26 are configured to provide a narrow horizontal beam width and a wide vertical beam width. According to a preferred embodiment, the side scan elements have a rectangular shape. According to a particularly preferred embodiment, the rectangular shaped side scan elements 26 are between about 3 inches to about 7 inches long by between about 0.125 inch and about 0.50 inch wide. According to a particularly preferred embodiment, the rectangular side scan elements 26 are about 4.5 inches long and about 0.25 inch wide. In such a particular preferred embodiment shown schematically in FIG. 1, the side scan elements 26 transmit the acoustic beam 47 with a horizontal angle 49 of about 2 degrees and a vertical angle 51 of about 50 degrees.

[0040] Referring to FIGS. 1, 4, 6, and 7, the side scan elements 26 are supported (e.g., captured, cradled, secured, etc.) by projections 42 in housing 24 so that their exposed

surface **48** is orientated at a predetermined direction and angle. According to an exemplary embodiment, the side scan elements **26** are supported by the housing so that the exposed surface is orientated outward from the transducer **20** and toward the bottom of the water (e.g., downward from the watercraft **10** and surface of the water). According to a preferred embodiment, side scan elements **26** are angled downward between about 20 degrees and about 40 degrees, depending on the resonant frequencies. According to a particularly preferred embodiment, the side scan elements **26** are orientated downward at about 30 degrees for resonant frequencies of between about 260 KHz and about 462 KHz. According to alternative embodiments, the side scan elements are mounted with any of a variety of orientations and directions, depending on types of depths the transducer is intended to be used in (e.g., lake, river, ocean, etc.) and on the configuration of the transducer sonar beams (e.g., as determined by the size and dimensions of the acoustic elements). Preferably, the side scan elements **26** are coupled to the housing **24** with an epoxy. Alternatively, the side scan elements are coupled to the housing by any of a variety of adhesives or bonding or joining materials or techniques.

[0041] According to a preferred embodiment, side scan elements **26** are made from a piezoelectric ceramic. According to a particularly preferred embodiment, the side scan elements are composed of lead zirconate titanate ("PZT") commercially available from Morgan Electro Ceramics of the United Kingdom. According to alternative embodiments, the side scan elements may be made from any of a variety of piezoelectric materials capable of converting electric energy into mechanical energy and converting mechanical energy into electrical energy.

[0042] An acoustic shield **50** (e.g., shielding, decoupler, barrier, absorber, etc.) surrounds all but one side of side scan elements **26** to prevent sonar pulses from being transmitted, and sonar returns received by, acoustic elements in all but the desired direction of scanning. The acoustic shield **50** may be made from any of a variety of materials that are poor conductor of sonar energy, such as cork, foam, polymers, or other low density materials, and the like.

[0043] The downward scan element **28** is located along the bottom middle of the housing **24** and is configured to scan the water below the transducer **20** (and watercraft **10**) with sonar or acoustic beams **53** (see FIG. 1). According to a preferred embodiment, downward scan element **28** comprises a pair of transducer elements coupled together. According to alternative embodiments, the downward scan element comprises a single element or more than two elements.

[0044] The dimensions of downward scan elements **28** are configured to provide a desired sonar beam pulse. According to an exemplary embodiment, the downward scan element **28** is configured to provide a relatively narrow sonar beam (e.g., for desired or optimum resolution). According to a preferred embodiment, the downward scan elements **28** have a cylindrical shape. According to a particularly preferred embodiment, the cylindrical shaped downward scan elements **28** have a diameter of between about 1 inch to about 2 inches and a height of between about 0.2 inches and about 0.5 inches. According to a particularly preferred embodiment, the cylindrical downward scan element **28** has a diameter of about 1.67 inches and a height of about 0.425 inch. In such a particular preferred embodiment shown in FIG. 1, the downward scan elements **28** transmit the acoustic beam **53** with an angle **55** of about 20 degrees. According to alternative embodiments, the

side scan elements and the downward scan elements may have any of a variety of dimensions, positions, and orientations based on desired performance, manufacturing, and costs.

[0045] The downward scan elements **28** are supported (e.g., captured, cradled, secured, etc.) by projections **52** in housing **24** so that its exposed surface **54** is orientated at a predetermined direction and angle. According to an exemplary embodiment, the downward scan elements **28** are supported by the housing so that the exposed surface is orientated vertically downward from the transducer **20** and toward the bottom of the water. Preferably, the downward scan elements **28** are coupled to the housing **24** with an epoxy. Alternatively, the downward scan elements are coupled to the housing by any of a variety of adhesives or bonding or joining materials or techniques.

[0046] According to a preferred embodiment, the downward scan elements **28** are made from a ceramic. According to a particularly preferred embodiment, the downward scan elements **28** are composed of lead zirconate titanate ("PZT") commercially available from Morgan Electro Ceramics of the United Kingdom. According to alternative embodiments, the downward scan elements may be made from any of a variety of piezoelectric materials capable of converting electric energy into mechanical energy and converting mechanical energy into electrical energy.

[0047] An acoustic shield **56** (e.g., shielding, decoupler, barrier, absorber, etc.) surrounds all but one side of the downward scan elements **28** to prevent sonar pulses from being transmitted, and sonar returns received by, acoustic elements in all but the desired direction of scanning. The acoustic shield **56** may be made from any of a variety of materials that are poor conductor of sonar energy, such as cork, foam, polymers, or other low density materials, and the like.

[0048] The return sonar signal from the bottom reflection carries details about the bottom **16**. The return sonar signal from the side reflection carries details about the sides and bottom **16** to the side of the watercraft **10**. The sonar return data is communicated or sent to be processed by the transducer **20** to the electronic control head unit **22** for display of images or symbols representative of the received return echoes of the acoustic wave beams. The transmission of an acoustic pulse and the reception of reflected echoes is a transmit/receive cycle, which is referred to herein as a T/R cycle. The wavefront of the acoustic pulse travels from the transducer **20**, to the bottom **16** of the body of water **14**, and reflects back to the transducer which receives the reflected echoes of the acoustic wave beam. The duration of the T/R cycle depends on the depth of the water. Typically, the T/R cycles of transmission and reception are two to four times per second for deep water and more frequently, such as one-thirtieth of a second, for shallower waters.

[0049] According to a preferred embodiment, the transducer **20** does not include any electronics; rather the electronics are located in the electronic control head unit **22**. The images include a bottom profile, objects along the bottom or in the water (e.g., fish), and the like. The display may also display informational subject matter (e.g., depth, water temperature, velocity of the watercraft **10**, etc.).

[0050] Referring to FIG. 12, the electronic control head unit **22** is coupled to a power supply and comprises a user interface (**58**, **60** and **62**), a microprocessor **64**, a co-processor **66**, a first side scan circuit **68**, a second side scan circuit **70**, and a bottom scan circuit **72**. The user interface is configured

to allow for user inputs through a display menu where parameters like depth range, sensitivity, fish alarm and the like. The user interface is shown to comprise a keypad **58**, buzzer **60**, and display **62**. Alternatively, the user interface may have switches or push buttons, or the like.

[0051] The microprocessor **64** is coupled to the user interface and is configured to process the data from the co-processor **66** (e.g., control the displayed information, format the information for display, run the operational algorithms, and the like). The microprocessor **64** can be a microcontroller, application-specific integrated circuit (ASIC) or other digital and/or analog circuitry configured to perform various input/output, control, analysis, and other functions described herein. In one embodiment, the micro-processor **64** includes a memory (e.g., non-volatile memory) configurable with software to perform the functions disclosed herein. The microprocessor **64** of the electronic control head unit **22** implements programmed algorithms (e.g., differential amplitude filtering (eliminate engine spark noise), time variable gain optimization-for best image, fish finding algorithms, anti-ranging pulse on transmit for better resolution, and use down beam depth to correct slant angle range information). According to a preferred embodiment, a software filter algorithm is provided to filter certain noise common to operation of watercraft (and noise caused by sparkplug in particular).

[0052] During operation of the sonar imaging system **18** in a particularly preferred embodiment, amplitude readings are taken approximately every 0.75 inches, such that 100 feet of depth has 1600 readings. The 0.75 inch amplitude readings from the last transmit/receive cycle (T/R cycle) are saved into computer memory. For each of these 0.75 inch amplitude readings, present and previous amplitude readings the software conducts the following test: Is “present reading”-“previous reading”>x. If Yes, then substitute “previous reading” for “present reading”. If No, use the present reading. The microprocessor **64** also filters the signals, sorts sonar target returns from the bottom and fish, calculates display range parameters and then feeds the processed signals to the LCD display screen. The display **62** is preferably a graphic display, for example, but not limited on the pixel order. Other displays such as LED, flasher, A-scope and digital segment may alternatively be used. The electronic control head unit **22** may be powered by batteries (e.g., its own dedicated batteries, marine battery, etc.).

[0053] The co-processor **66** is coupled to the microprocessor **64** and is configured to collect, process, and pass data to the microprocessor **64** (e.g. generates the transmission frequencies, converts the analog data to digital with A/D converter and sends to the microprocessor **64**). The co-processor **66** can be a microcontroller, application-specific integrated circuit (ASIC) or other digital and/or analog circuitry configured to perform the functions disclosed herein. In one embodiment, the co-processor **66** includes a memory (e.g., non-volatile memory) configurable with software to perform the functions disclosed herein.

[0054] The first side scan circuit **68** is coupled to the co-processor **66** and is configured to operate one of the side scan elements. The first side scan circuit **68** comprises a receiver **74**, a transmitter **76**, and a transmit/receive switch (i.e., T/R switch **78**). The second side scan circuit **70** is coupled to the co-processor **66** and is configured to operate the other side scan element. The second side scan circuit **70** comprises a receiver **86**, a transmitter **88**, and a transmit/receive switch (i.e., T/R switch **90**). The bottom scan circuit **72** is coupled to

the co-processor **66** and is configured to operate the bottom scan element. The bottom scan circuit **72** comprises a receiver **80**, a transmitter **82**, and a transmit/receive switch (i.e., T/R switch **84**).

[0055] The receivers **74**, **80**, **86** are configured to amplify the signal and conducts signal filtering, base banding-rectification (e.g., remove carrier frequency), and logarithmic conversions (e.g., to obtain a wide range at output) and preferably provide variable receiver bandwidth. The transmitters **76**, **82**, **88** are configured to drive the acoustic elements and preferably provide variable transmit power and preferably at a high voltage. The T/R switches **78**, **84**, **90** are configured to switch the first side scan circuit **68** between transmit and receive modes.

[0056] According to a preferred embodiment, the electronic control head unit **22** is configured to operate at one or more resonant frequencies, depending on the intended depth and desired resolution. Such a multiple-frequency operation is intended to make up for shortcomings of mounting the transducer to the watercraft **10** caused by the varying distance between the transducer to the bottom **16** of the water **14**. According to a particularly preferred embodiment using a dual-resonant frequency and side scan acoustic elements that are about 4.5 by about 0.25 inch, the electronic control head unit **22** is configured to operate at 260 kHz resonant frequency (e.g., wider acoustic wave beam for deeper depth and further distances) and at 462 kHz resonant frequency (e.g., narrower acoustic wave beam for shallower depth and shorter distances). The down beam that is provided in one embodiment utilizes a 200 kHz/150 kHz dual beam. Other embodiments may use a quad beam or even six beam. In preferred embodiments, at least one view of the display shows both the down beam imaging and side imaging. This provides the ability to better relate length of shadow information to the size of the underwater target.

[0057] FIG. 1 shows a cross-sectional view of the body of water **14** to illustrate features of the present invention during operation of the sonar imaging system **18**. With additional reference to FIG. 12, the sonar imaging system **18** transmits the acoustic wave beam **47** from the side scan elements **26** and the acoustic wave beam **53** from downward scan elements **28**. The receivers **74**, **80**, **86** begin listening for sonar returns through the transducer **20**. The acoustic wave beams **47**, **53** propagate to the bottom surface **16** and reflects a sonar return. The transducer **20** communicates the received sonar return to the receivers **74**, **80**, **86**. A prior cycle had determined the depth, and in the illustrated embodiment, the depth is displayed on the display **62** as well as provided to a controller for evaluating the duration. Using the prior determined depth, a controller determines an approximate travel time for the sound energy signal **47**, **53** to reach the bottom **16** and return. At a predetermined proportion of the travel time, the return sonar reaches a point near the transducer. The return sonar from the bottom **16** reflection carries details about the bottom **16**. The controller directs the switch to change the receiving mode from transmit mode to receive mode. The receivers **74**, **80**, **86** then use the acoustic elements for the return sonar. The sonar imaging system **18** continues receiving in the narrow acoustic wave beam mode, until the start of the next T/R cycle. The received sonar returns are processed by the controller for display of representative symbols on the display **62**. The T/R cycle then repeats with the newly determined depth from the prior cycle.

[0058] The sonar images from the down beam and side scan elements are then displayed on the display **62**. These images may be shown in grey-scale or in color. The location of the watercraft **10** is also shown in the image. If the user chooses to only display the down beam sonar information, historic information is typically shown to the left of the location of the watercraft. As such, the display **62** shows images to the bottom of the watercraft **10** that are even with and behind the watercraft **10** when the watercraft **10** is traveling forward. The user may also display only the side scan sonar images, only those from one side scan element, or images from both sides and the bottom. The display **62** may also be configured (or configurable) to indicate information such as depth, and speed of the watercraft **10**, range, etc.

[0059] In a highly preferred embodiment, a GPS receiver **92** is also included to provide the microprocessor **64** location information. This information may be used to provide charting and other navigational functions. To provide even more accurate images, the system of the present invention provides the offset necessary to account for the X and Y distance between the side imaging transducer and the GPS antenna. In one embodiment a cursor mode allows a user to move a cursor on the display **62** over a target of interest on the screen image and set a waypoint for the location of the structure. The GPS history may be used to determine the distance back and the sonar may be used to determine the distance to the side. The GPS speed is used in one embodiment to provide the screen scroll rate to provide more accurate front to back target dimensions. Without GPS or a speed sensor a fast scroll rate and a slow boat speed will elongate targets and a slow scroll rate and a fast boat speed will shorten targets. The corners of screen captures can be marked so that large area composite mosaic images can be generated in the unit **22** or post processed later. Preferably, one view that shows both side imaging and navigation information is provided. This makes it easier to follow tracks and provide efficient area coverage.

[0060] Referring to FIGS. **13-18**, exemplary sonar images **92** reproduced from a display **62** are shown in grey-scale. Alternatively, the display **62** may display color sonar images. The location of the watercraft **10** is shown as "0" in the images **92**. Historic information is shown to the left of "0". As such, the display **62** shows images **92** to the bottom or side of the watercraft **10** that are even and behind the watercraft **10**. A range **105** to the side of the watercraft **10** is also provided. A dark area **106** along the top of the display **62** shows the water column above the bottom **16**. The portion **93** of the image **92** below the water column dark area **106** shows returns from the bottom **16**. As shown by the returns in the sonar images **92**, the body of water **14** includes the bottom **16** and underwater articles such as trees **94**, logs **96**, a sunken barge **98**, and a school of fish **100**. Shadows **102** from the underwater articles indicate the general distance or displacement of the underwater article (e.g., distance it extends from the bottom **1**). Dark areas **104** typically indicate drop offs, holes, and the like. The display **62** may also be configured (or configurable) to indicate information such as depth **108**, and speed **110** of the watercraft **10**.

[0061] According to a preferred embodiment shown in FIG. **19**, the display **62** is configurable to provide an image **120** that shows the underwater environment using both side scan elements (views) **26** and the downward scan element **28**. An icon **122** (in the form of a schematic watercraft) indicates the position and orientation and direction of travel of the watercraft **10**.

[0062] It is important to note that the terms are intended to be broad terms and not terms of limitation. These components may be used with any of a variety of products or arrangements and are not intended to be limited to use with fish finding applications. For example, mounting to a watercraft is not intended to be limiting to devices that are directly attached to the watercraft, but would include devices attached to motors (such as trolling motors) attached to the watercraft, and the like.

[0063] It is also important to note that the construction and arrangement of the elements of the sonar imaging system for mounting to a watercraft as shown in the preferred and other exemplary embodiments are illustrative only. Although only a few embodiments of the present invention have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited in the claims. For example, the transducer preferably provides dual frequency, single element side beams in the form of two opposed vertical beams optimized for range and depth and front to back beam width selected based on image resolution, fish finding and transducer length.

[0064] Elements shown as integrally formed may be constructed of multiple parts or elements show as multiple parts may be integrally formed, the operation of the interfaces may be modified or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied (e.g. by variations in the number of engagement slots or size of the engagement slots or type of engagement). It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures and combinations. Accordingly, all such modifications are intended to be included within the scope of the present invention as defined in the appended claims.

[0065] The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. In the claims, any means-plus function clause is intended to cover the structures described herein as performing the recited function and not only structural equivalents but also equivalent structures. Other substitutions, modifications, changes and/or omissions may be made in the design, operating conditions and arrangement of the preferred and other exemplary embodiments without departing from the spirit of the present invention as expressed in the appended claims.

[0066] All references, including publications, patent applications, and patents cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0067] The use of the terms "a" and "an" and "the" and similar referents in the context of describing the invention (especially in the context of the following claims) is to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms "comprising," "having," "including," and "containing" are to be construed as open-ended terms (i.e.,

meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

[0068] Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

1. A sonar system mounted to a boat to provide detailed images of underwater articles located at the left and right sides of the boat, the sonar system comprising:

left and right side scan elements for transmitting sonar beams directed downward from the boat and, respectively, outward to the left and right sides of the boat, the sonar beams having a narrow horizontal beam width and a wide vertical beam width, and for receiving detailed sonar returns reflected from and providing details of the underwater articles;

signal processing circuitry for processing the detailed sonar returns to produce digital data;

data processing circuitry for processing the digital data to produce signals corresponding to the detailed images of the underwater articles; and

a display for receiving the signals and displaying the detailed images of the underwater articles located at least one of the left and right sides of the boat.

2. The system of claim 1, wherein the display displays left and right side detailed images of underwater articles respectively located at the left and right sides of the boat.

3. The system of claim 1, wherein the detailed sonar returns further provide shadows from the underwater articles, and the detailed images of the underwater articles include these shadows.

4. The system of claim 1, wherein the detailed sonar returns further provide underwater dark areas, and the detailed images of the underwater articles include the underwater dark areas.

5. The system of claim 1, wherein the display further displays a water depth.

6. The system of claim 1, wherein the display further displays a speed of the boat.

7. The system of claim 1, wherein the detailed images further show the bottom of a body of water.

8. The system of claim 1, wherein the detailed images are shown in grey-scale or color.

9. The system of claim 1, wherein the display is part of a user interface configured to allow for user inputs through a display menu.

10. The system of claim 1, wherein the left and right side scan elements are orientated downward from the surface of the water at an angle between about 20 degrees and about 40 degrees.

11. The system of claim 1, wherein the left and right side scan elements operate at resonant frequencies of between about 260 KHz and about 462 KHz.

12. The system of claim 1, wherein a cross-section of each of the left and right side scan elements is a rectangle having a length of up to about 7 inches and a width of up to about 0.50 inch.

13. The system of claim 1, where the left and right side scan elements are made from a piezoelectric ceramic.

14. The system of claim 1, further comprising an acoustic shield which substantially surrounds each of the left and right side scan elements.

15. The system of claim 1, further comprising a downward scan element for transmitting a sonar beam directed straight down from the boat.

16. A sonar system to provide detailed images of underwater articles located at the left and right sides of a boat, the sonar system comprising:

a transducer assembly comprising:

a housing having a mounting member that is mounted to the boat; and

left and right side scan elements positioned within the housing for transmitting sonar beams directed downward from the boat and, respectively, outward to the left and right sides of the boat, the sonar beams having a narrow horizontal beam width and a wide vertical beam width, and for receiving detailed sonar returns reflected from and providing details of the underwater articles;

signal processing circuitry for processing the detailed sonar returns to produce digital data;

data processing circuitry for processing the digital data to produce signals corresponding to the detailed images of the underwater articles; and

a display for receiving the signals and displaying the detailed images of the underwater articles located at least one of the left and right sides of the boat.

17. The system of claim 16, wherein the display displays left and right detailed images of underwater articles located at the left and right sides of the boat.

18. The system of claim 16, wherein the left and right side scan elements are respectively positioned within the housing at a depression angle of between about 20 degrees and about 40 degrees.

19. The system of claim 18, wherein the left and right side scan elements are respectively positioned at their depression angles via a cradle.

20. The system of claim 16, wherein the signal processing circuitry, data processing circuitry and display are assembled within an electronic control head unit.

21. The system of claim 20, wherein the electronic control head unit and the left and right side scan elements of the transducer assembly are operatively coupled by a cable.

22. The system of claim 16, wherein the housing comprises a top housing portion and a bottom housing portion.

23. The system of claim 22, wherein the mounting member extends from the top housing portion.

24. The system of claim 16, wherein the left and right side scan elements operate at resonant frequencies of between about 260 KHz and about 462 KHz.

25. The system of claim 16, wherein a cross-section of each of the left and right side scan elements is a rectangle having a length of up to about 7 inches and a width of up to about 0.50 inch.

26. A method of providing detailed images of underwater articles located at the left and right sides of a boat using a sonar system mounted to the boat, the method comprising the steps of:

transmitting sonar beams directed downward from the boat and outward to the left and right sides of the boat, the sonar beams having a narrow horizontal beam width and a wide vertical beam width;

receiving detailed sonar returns reflected from and providing details of the underwater articles;

processing the detailed sonar returns to produce digital data;

processing the digital data to produce signals corresponding to the detailed images of the underwater articles; and processing the signals to display the detailed images of the underwater articles located at least one of the left and right sides of the boat.

27. A sonar system mounted to a boat to provide detailed images of underwater articles located at the left and right sides of the boat, the system comprising:

a left side scan sonar transducer for transmitting a left side scan sonar beam of pulses directed downward and laterally to the left side of the boat, the left side scan sonar beam having a narrow horizontal beam width and a wide vertical beam, and for receiving left side scan sonar return signals from the underwater articles;

a right side scan sonar transducer for transmitting a right side scan sonar beam of pulses directed downward and laterally to the right side of the boat, the right side scan sonar beam having a narrow horizontal beam width and a wide vertical beam width, and for receiving right side scan sonar return signals from the underwater articles;

signal processing circuitry for processing the left and right side scan sonar return signals to produce side scan image data;

a display; and

a digital processor for providing signals to the display based upon the side scan image data to produce on the display a detailed image showing details and shadows of one or more underwater articles located at least one of the left and right sides of the boat, wherein the detailed image is shown in grey-scale or color.

28. The system of claim 27, wherein in response to a user input via a touch screen or keypad, the digital processor causes the display to zoom into a particular area of the detailed image.

29. The system of claim 27, wherein the display displays left and right side detailed images of underwater articles respectively located at the left and right sides of the boat.

30. The system of claim 29, wherein a water column between the boat and the bottom of the body of water is displayed between the left and right side detailed images.

31. The system of claim 27, and further comprising:

a down beam sonar transducer for producing a sonar beam of pulses directed downward of the boat, and for receiving down sonar return signals from the underwater articles or the bottom of the body of water.

32. The system of claim 31, wherein the digital processor provides signals to cause the display to show a numerical representation of the depth of the bottom of the body of water based upon the down sonar return signals.

33. The system of claim 31, wherein the signal processing circuitry produces down sonar data based upon the down sonar return signals, and wherein the digital processor provides signals to the display based upon the down sonar data.

34. The system of claim 33, wherein a view based upon the down sonar data is displayed on the display.

35. The system of claim 27, wherein each of the left and right side scan sonar transducers is operable at a first frequency and a second frequency to provide detailed images of varying resolution.

36. The system of claim 27, wherein a cross-section of each of the left and right side scan elements is a rectangle having a length of up to about 7 inches and a width of up to about 0.50 inch.

37. A sonar system comprising:

a GPS receiver for providing GPS position data;

a sonar transducer assembly mounted to a boat, the sonar transducer assembly including a left side scan sonar transducer, a right side scan sonar transducer, and a down beam sonar transducer;

signal processing circuitry for processing sonar return signals from the left side scan sonar transducer, the right side scan sonar transducer, and the down beam sonar transducer;

a user interface including a touch screen display or keypad for receiving a first user input; and

a digital processor for controlling the display based upon the processed sonar return signals and the GPS position data, to display information in a mode selected by the first user input from a group consisting of one or more modes:

a mode in which the digital processor causes the display to display a grey-scale or color image that provides details and shadows of one or more underwater articles based upon sonar return signals received by one of the left and right side scan sonar transducers;

a mode in which the digital processor causes the display to simultaneously show a grey-scale or color image that provides details and shadows of one or more underwater articles based upon sonar return signals received by both of the left side scan sonar transducer and the right side scan sonar transducer;

a mode in which the digital processor causes the display to show information based upon the GPS position data; and

a mode in which the digital processor causes the display to simultaneously show a grey-scale or color image that provides details and shadows of one or more underwater articles based upon sonar return signals received by one or both of the left side scan transducer and the right side scan transducer, and information based upon GPS position data.

38. The system of claim 37, wherein the digital processor causes the display to zoom into a particular area of the grey-scale or color image in response to a second user input.

39. A method of providing detailed images of underwater articles located at the left and right sides of a boat using a sonar system mounted to the boat, the method comprising the steps of:

transmitting left and right side scan sonar beams of pulses with left and right side scan sonar transducers, respectively, wherein the left and right side scan sonar transducers and the down beam sonar transducer are mounted to the boat, and wherein the left side scan sonar beam is directed downward and laterally to the left side of the boat, the left side scan sonar beam having a narrow horizontal beam width from the front to the back of the boat and a wide vertical beam width in a plane perpendicular thereto, and wherein the right side scan sonar beam is directed downward and laterally to the right side of the boat, the right side scan sonar beam having a narrow horizontal beam width from the front to the back of the boat and a wide vertical beam width in the plane perpendicular thereto;

receiving left and right side scan sonar return signals with the left and right side scan sonar transducers, respectively;

generating side scan image data from the left and right side scan sonar return signals;

displaying a detailed image that, based upon the side scan image data, provides details and shadows of one or more underwater articles located at least one of the left and right sides of the boat, wherein the detailed image is shown in grey-scale or color.

40. The method of claim **39**, and further comprising:

transmitting a sonar beam of pulses with a down beam sonar transducer;

receiving down sonar return signals with the down beam sonar transducer; and

generating and displaying a numerical representation of the depth of the bottom of the body of water based upon down sonar return signals.

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